CHAPTER 11

THERMODYNAMICS

CHAPTER ESSENCE AT A GLANCE:
Governs heat and work transformations, the First Law of Thermodynamics, isothermal and adiabatic processes, Mayer's relation, and the Second Law of Thermodynamics.

4 CORE PRINCIPLES TO UNDERSTAND FIRST

• **State Variables: Intensive vs Extensive:** Intensive variables do NOT depend on system size (Temperature, Pressure, Density). Extensive variables ARE proportional to size (Volume, Internal Energy U, Mass).

• **First Law of Thermodynamics:** ΔQ = ΔU + ΔW (Conservation of Energy). Work done by gas ΔW = P ΔV (area under P-V diagram). For cyclic process, ΔU = 0 implies Q = W.

• **4 Thermodynamic Processes:** Isothermal (T = const, ΔU = 0, W = nRT ln(V_2 divided by V_1)); Adiabatic (Q = 0, PV^γ = const, W = nR(T_1-T_2) divided by (γ-1)); Isochoric (V = const, W = 0); Isobaric (P = const, W = PΔV).

• **Mayer's Relation:** C p - C v = R. At constant pressure, heat must both increase internal energy AND perform external expansion work (P ΔV = R ΔT).

ESSENTIAL FORMULAS & GOVERNING EQUATIONS

$$\Delta U = q + w = q - P\Delta V$$

First Law of Thermodynamics

$$C_p - C_v = R \quad (\text{Mayer's Relation})$$

Molar Heat Capacities at Constant P and V

$$W_{\text{iso}} = nRT\ln\left(\frac{V_2}{V_1}\right) \quad (\Delta U = 0)$$

Work Done in Isothermal Expansion

$$W_{\text{adia}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}$$

Work Done in Adiabatic Expansion

CRUCIAL EXAM PRECAUTIONS & VALUATION RUBRIC SECRETS

• **Alert:** In an isothermal process, temperature is constant so ΔU = 0! In an adiabatic process, heat exchange is zero (Q = 0). Do not mix them up!

• **Alert:** Always state standard SI units, vector directions, and sign conventions in step calculations.

HIGH-YIELD CENTUM QUESTIONS & STEP-BY-STEP EXPLANATIONS

1. Define: (a) Isothermal process (b) Adiabatic process (c) Isobaric process (d) Isochoric process. [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Four Processes]** 1. Isothermal: Temperature remains constant (ΔT = 0, P V = const). 2. Adiabatic: No heat enters or leaves the system (ΔQ = 0, P V^γ = const). 3. Isobaric: Pressure remains constant (ΔP = 0, W = P ΔV). 4. Isochoric: Volume remains constant (ΔV = 0, W = 0). [2 Marks]

• **Key Exam Result:** Isothermal (T=const), Adiabatic (Q=const), Isobaric (P=const), Isochoric (V=const)

2. State the Second Law of Thermodynamics (Kelvin-Planck and Clausius statements). [Score: 2 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Kelvin-Planck Statement]** 1. Kelvin-Planck: It is impossible to construct a heat engine operating in a cycle that absorbs heat from a reservoir and converts it completely into mechanical work without rejecting any heat to a cooler sink (no heat engine has 100% efficiency). [1 Mark]

• **[Step 2: Clausius Statement]** 2. Clausius: It is impossible to construct a cyclic device that transfers heat from a colder body to a hotter body without any external work input (a refrigerator requires work). [1 Mark]

• **Key Exam Result:** Kelvin-Planck (No 100% engine), Clausius (Heat cannot flow cold -> hot without work)

3. Derive Mayer's relation: C p - C v = R for one mole of an ideal gas. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Heat at Constant Volume]** For 1 mole of an ideal gas at constant volume, heat supplied for temperature rise ΔT is: Δ Q v = C v ΔT. Since ΔV = 0, ΔW = 0 => ΔU = C v ΔT ---(1). [1 Mark]

• **[Step 2: Heat at Constant Pressure]** At constant pressure: Δ Q p = C p ΔT. By First Law: Δ Q p = ΔU + P ΔV => C p ΔT = C v ΔT + P ΔV ---(2). [1 Mark]

• **[Step 3: Ideal Gas Law Substitution]** For 1 mole of an ideal gas, P V = R T. At constant pressure: P ΔV = R ΔT. Substituting into (2): C p ΔT = C v ΔT + R ΔT => C p = C v + R => C p - C v = R. [1 Mark]

• **Key Exam Result:** C p - C v = R (Mayer's Relation)

4. Derive an expression for the work done during an isothermal expansion of n moles of an ideal gas from volume V1 to V2 at temperature T. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: Work Definition]** Work done during volume change dV is dW = P dV. Total work: W = $\int_{V1}^{V2}$ P dV. [1 Mark]

• **[Step 2: Substitution from Ideal Gas Equation]** For n moles of an ideal gas: P V = n R T => P = n R T divided by V. W = $\int_{V1}^{V2}$ (n R T divided by V) dV. Since T is constant in an isothermal process: W = n R T $\int_{V1}^{V2}$ (1 divided by V) dV. [1 Mark]

• **[Step 3: Integration & Log Form]** W = n R T [ln(V)] {V1}^{V2} = n R T ln(V2 divided by V1) = 2.303 n R T log10(V2 divided by V1). Since P1 V1 = P2 V2 ⇒ V2 divided by V1 = P1 divided by P2: W = 2.303 n R T log10(P1 divided by P2). [1 Mark]

• **Key Exam Result:** W = n R T ln(V2 divided by V1) = 2.303 n R T log10(V2 divided by V1)

5. Derive an expression for the work done during an adiabatic expansion of n moles of an ideal gas from temperature T1 to T2. [Score: 3 Marks]

Answer & Key Valuation Steps:

• **[Step 1: First Law for Adiabatic Process]** In an adiabatic process, ΔQ = 0. By First Law: ΔQ = ΔU + ΔW => 0 = ΔU + W => W = -ΔU. [1 Mark]

• **[Step 2: Internal Energy Relation]** For n moles of ideal gas: ΔU = n C v ΔT = n C v (T2 - T1). W = -n C v (T2 - T1) = n C v (T1 - T2). [1 Mark]

• **[Step 3: Expressing in terms of R and γ]** Since C p - C v = R and γ = C p divided by C v => C v = R divided by (γ - 1). Substituting: W = [n R (T1 - T2)] divided by (γ - 1) = (P1 V1 - P2 V2) divided by (γ - 1). [1 Mark]

• **Key Exam Result:** W adiabatic = [n R (T1 - T2)] divided by (γ - 1)

6. Calculate the work done when 1 mole of an ideal gas expands isothermally at 300 K to double its initial volume. (R = 8.31 J mol^{-1} K^{-1}, ln 2 = 0.693) **[Score: 3 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Formula]** For isothermal expansion: W = n R T ln(V2 divided by V1). [1 Mark]

• **[Step 2: Given Data]** n = 1 mole, T = 300 K, V2 divided by V1 = 2, R = 8.31 J mol^{-1} K^{-1}. [0.5 Mark]

• **[Step 3: Calculation]** W = 1 × 8.31 × 300 x ln(2) = 2493 × 0.693 = 1727.65 J approx. 1.73 kJ. [1.5 Marks]

• **Key Exam Result:** Work Done = 1727.65 J (1.73 kJ)

7. In an adiabatic process, a given mass of gas is compressed to half its initial volume. If the initial temperature was 27 °C, find the final temperature. (Take $\gamma = 1.4$ for the gas) **[Score: 4 Marks]**

Answer & Key Valuation Steps:

• **[Step 1: Adiabatic Temperature-Volume Relation]** T1 V1^(γ - 1) = T2 V2^(γ - 1) ⇒ T2 = T1 (V1 divided by V2)^(γ - 1). [1.5 Marks]

• **[Step 2: Data Substitution]** Initial temperature T1 = 27 + 273 = 300 K. Compression ratio V1 divided by V2 = 2. γ - 1 = 1.4 - 1 = 0.4. [1 Mark]

• **[Step 3: Numerical Evaluation]** T2 = 300 x $(2)^{0}$.4. Since 2^{0}.4 = 2^(2 divided by 5) ≈ 1.3195: T2 = 300 × 1.3195 = 395.85 K. In Celsius: t2 = 395.85 - 273 = 122.85 °C. [1.5 Marks]

• **Key Exam Result:** Final Temperature T2 = 395.85 K (122.85 °C)